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(B.pharm, MSc. in Health Supply Chain Management)

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Section 7

Inventory Management and inventory control system

WAREHOUSING AND DISTRIBUTION

- Warehousing is a key element of pharmaceutical supply chain management.
- It ensures the constant availability and flow of essential quality health commodities, in appropriate quantities, in a timely and cost-efficient manner.
- Warehouse efficiencies depend upon a combination of warehouse design, layout, infrastructure, systems, process and people.

Cont . . .

- Warehouse design element aims to maximize the utility of space, equipment, and efficiency of operations.
- Strategic factors affecting warehouse design
- **1. Product:** Understanding the characteristics of the products.
- **2. Space:** The dimensions and characteristics of the warehouse infrastructure play a huge role in deciding the layout structure inside a warehouse.
- These features impact the design of shelves, capacity of the installations and the distribution of the goods inside the warehouse. This should adhere to the requirement of building regulations.

Cont . . .

- **3. Equipment:** Racks, lifts or MHE should be considered in design proposal.
- **4. Circulation of goods:** Circulation here refers to the movement that happens in logistical operations, such as dispatch, receipt and order preparation processes.
- **5. Personnel:** The staff working in the warehouse is one of the most important assets. It is important to know the required number of employees, their level of training etc

Cont . . .

- The location of a warehouse should be situated on a flat ground.
- The flooring should be have weight bearing capacity to withstand the use to which they may be subjected, e.g. physical damage from lift trucks and wheeled equipment or corrosion from chemical substances.
- The ground should be flat, even and smooth surface to facilitate material handling equipment movements and dust free.
- The walls and roof should be designed with suitable lighting panels and ventilators for air exchange.

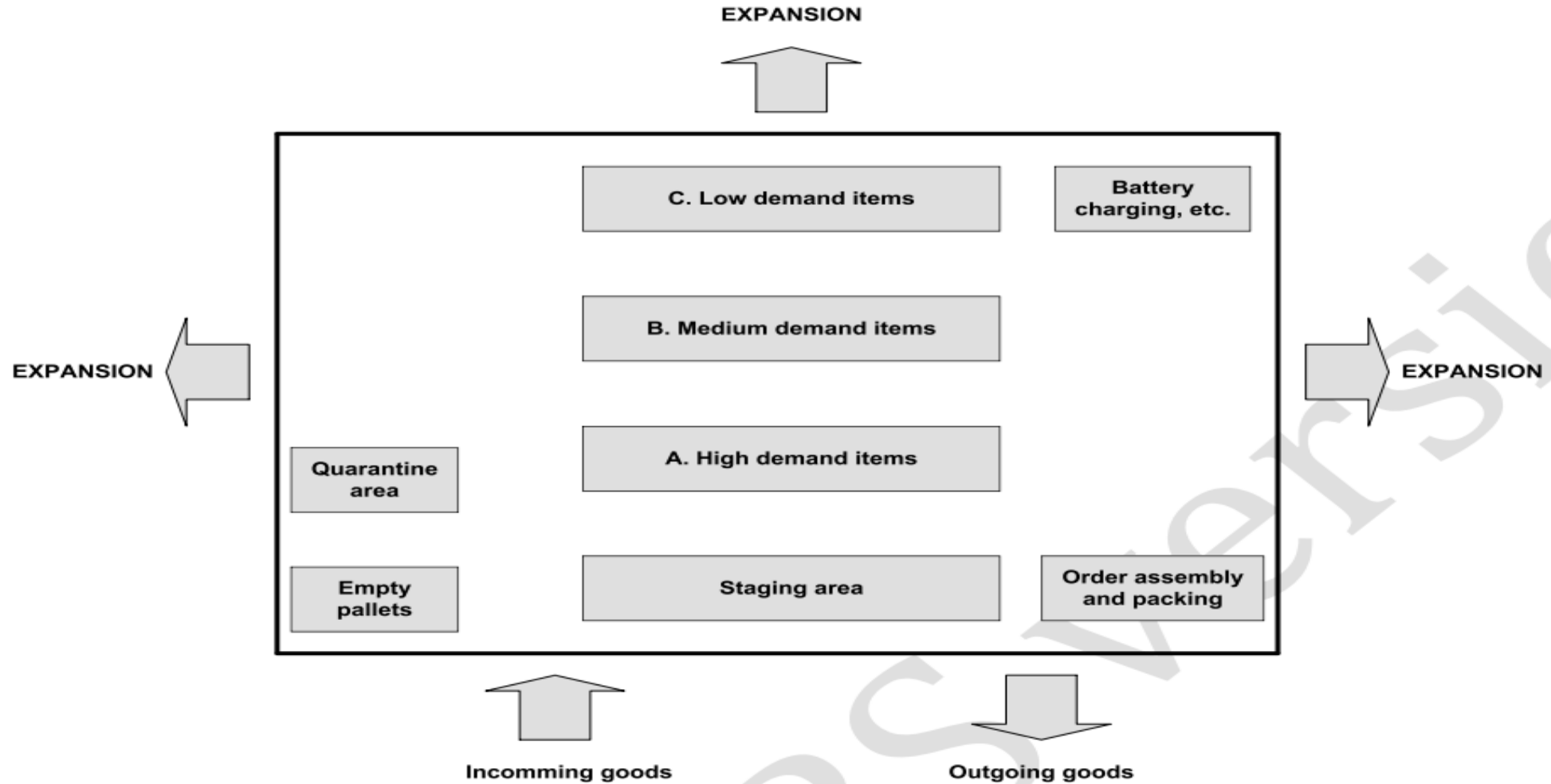
Cont . . .

- Internal layout
- Internal layout design should be built considering operational process, the nature of products, the volumes of products, storage types storage types, in-house operations involving put away and pull sequences.

Cont . . .

- There are two main layout options – the '*U*' flow the '*Through*' flow.
- Figure 1 shows the 'U' flow arrangement.
- **'U' Flow warehouse**
- Goods receipt and dispatch are located on the same side of the building and products are taken into stock in accordance with their ABC designation, with the highest demand items nearest the loading bays.

Figure 1 – ‘U’ flow warehouse



Adapted from Richards, 2011.

Cont . . .

Advantages of 'U' flow:

- Good utilization of dock resources because the receiving and shipping processes can share dock doors.
- Facilitates cross-docking because the receiving and dispatch areas are next to one another and can operate together.
- Excellent lift truck utilization because put away and retrieval trips are easily combined and storage locations closest to the receiving and dispatch docks are natural locations to house fast moving 'A'-rated items.
- Provides excellent security because only one side of the building is used for entry and exit.
- Allows scope for expansion in three directions.

Cont . . .

Disadvantage of the 'U' flow:

- Arrangement is that congestion can occur if there is heavy incoming and outgoing traffic at the same time.

'Through' flow warehouse

- Goods receipt and dispatch are located on opposite sides of the building. High demand items are stored along the central axis.

Advantage of through flow warehouse:

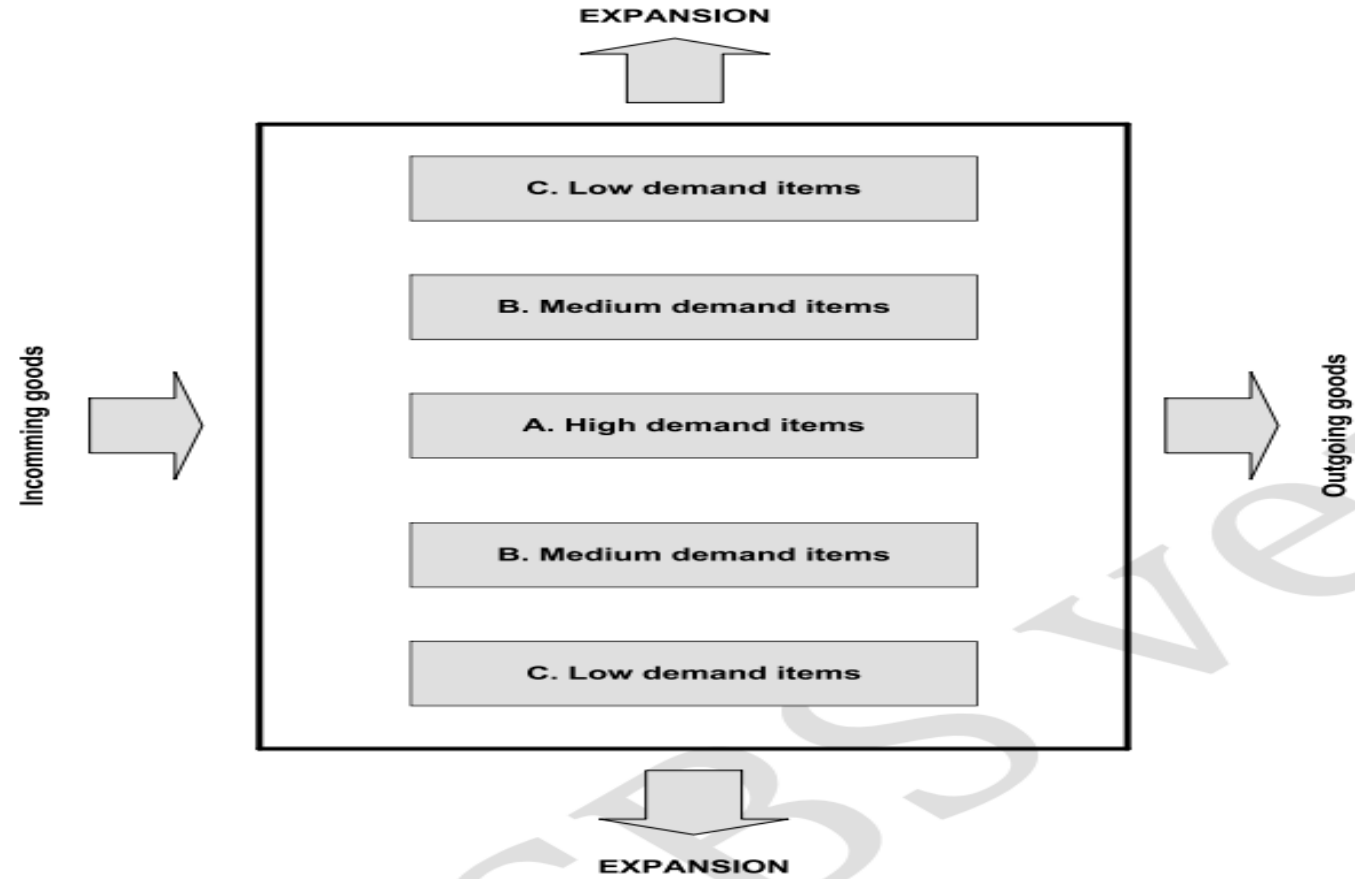
- Little risk of congestion at the loading docks.

Cont . . .

Disadvantage of the through flow warehouse:

- Security is an increased problem because of the two-sided arrangement. This is likely to require two security gates with access roads on both sides of the building. In addition, the potential for expansion is limited to two directions only.
- Figure 2 shows the 'Through' flow arrangement

Figure 2 – ‘Through’ flow warehouse



Adapted from Richards, 2011.

Good storage practice

- ✓Clean and disinfect storeroom regularly
- ✓Store supplies in a dry, well-lit, well-ventilated storeroom out of direct sunlight.
- ✓Secure storeroom from water penetration
- ✓Ensure that fire safety equipment is available and accessible and personnel are trained to use it
- ✓Store condoms and other latex products away from electric motors and fluorescent lights
- ✓Maintain cold storage, including a cold storage area for commodities that require it.
- ✓Keep narcotics and other controlled substances in a locked place.



GSP cont. . .



Store flammable products separately from other products. Take appropriate safety precautions.

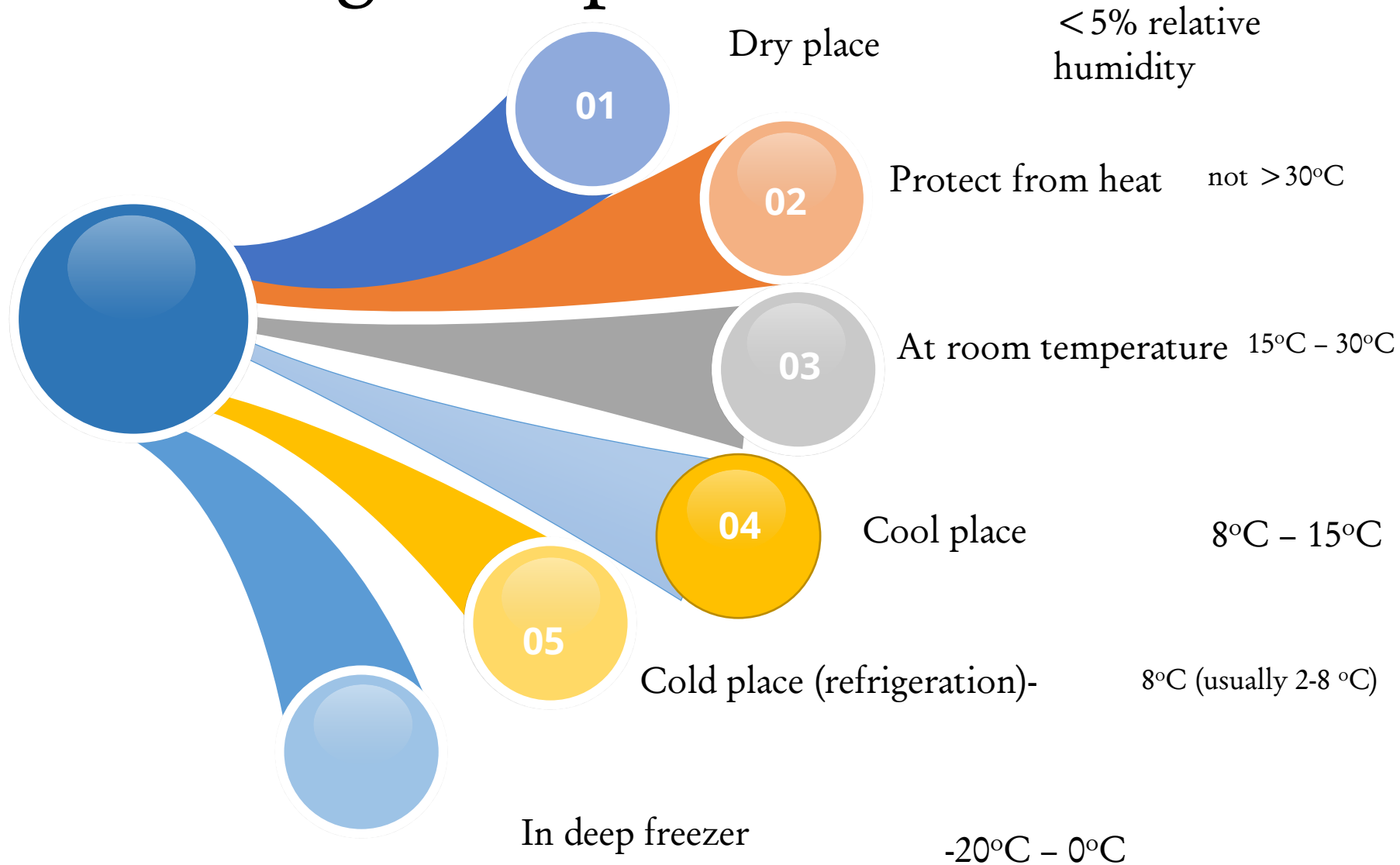
Stack cartons at least 10 cm (4 in) off the floor, 30 cm (1 ft) away from the walls and other stacks, and not more than 2.5 m (8 ft) high.



- Store medical supplies away from insecticides, chemicals, old files, office supplies, and other materials.
- Arrange cartons so that arrows point up. ensure that identification labels, expiry dates, and manufacturing dates are clearly visible.



Storage temperature and environment



Cont . . .

- Pharmaceutical warehousing or warehouse management is the physical movement of stock into, through and out of a medical store warehouse.
- Breaking it down, warehouse operations covers a number of important areas, these areas include:
 - ✓ Request
 - ✓ Receiving
 - ✓ Physical inspection
 - ✓ Storage
 - ✓ Stock arrangement
 - ✓ Issuing
 - ✓ Reverse logistics
 - ✓ Document management & filing

Cont . . .

Request

- A paper or electronic request form that may be used internally to requisition items from existing warehouse stores.
- The form must typically include an item number and/or description, quantity needed, date required, and the location of the individual or department making the request.
- A supply requisition must typically go through an approval process before being fulfilled.
- **Re-order(RO)** is the level of drugs in stock; it indicates when fresh orders should be made.
 - It is the quantity that is calculated to last between the period of placing the order and the delivery of the new consignment.
 - It should be updated at least twice a year because consumption may vary due to seasonal changes or epidemics.

Cont . . .

- **Physical inspection**
- **Visual inspection** is the process of examining products and their packaging by eye to look for problems in product quality.
- Should be completed each time products are handled: when receiving, issuing or dispensing supplies, or when conducting a physical count.
- When conducting a **visual inspection**, be sure to check the following:
 - **Package and product integrity:** check for damage to packaging (tears, perforations, water or oil) and products (unexpected odor, caking on bottles, change in color, broken bottles).
 - **Manufacturing defects:** incomplete supply, missing or illegible identification information.
 - **Labeling:** Make sure that products are labeled with the date of manufacture or expiration, lot number and manufacturer's name.

Cont . . .

- Stock arrangement
- Items can be physical organized in one of several ways:
 - A. Therapeutic or pharmaceutical category**
 - Most useful in small store rooms or dispensaries where the storekeeper is very knowledgeable about pharmacology.
 - **Purchased drugs** will be grouped following EFDA's Pharmacotherapeutic categorization, e.g. CNS drugs, Antinfectives, GIT,.....
 - Each Pharmacotherapeutic category will then further be arranged alphabetically and dosage form.

Cont . . .

- **Free drugs** will be grouped according to their program category, e.g. TB, Malaria, Family planning (FP), etc and each program category will then further be arranged alphabetically and dosage form.

B. Alphabetical order

- Is also attractive in peripheral stores that keep a small number of items. However, each change in the national (essential) medicines list or in the level of use list requires reorganization of the stores themselves and of the stock administration system.

Cont . . .

C. DOSAGE FORM

- A system based on dosage form is commonly used in smaller warehouse. Tablets and capsules are stored together, with separate areas for orally liquids, Injection, Creams and Ointments and Topical Liquids. Within each dosage form area, products may be stored in a fixed, fluid or semi-fluid manner and further organized by any of the other.

Cont . . .

.Stock rotation

- First Expiry First Out (FEFO):
 - Involves arranging drugs with the earliest expiration date in front/on top and drugs with the latest expiration date in the back/below so that **drugs with near expiry will be issued and/or dispensed first.**
- Other approaches: First In First Out(FIFO), Last In First Out (LIFO)

Cont . . .

- **Recording and reporting forms**

- From a logistics point of view, only three things can happen to supplies in a pipeline. They can be stored, moved (in transit), or consumed (used).
- Because we want to monitor products at all times in the pipeline, we need three types of logistics records to track the products. Each record type has a distinct form and use.

The three types of Recording and reporting forms

- **Stock keeping records-** Holds information about products in storage.
- **Transaction records-** Holds information about products being moved.
- **Consumption records-** Holds information about products being consumed or used.

Cont . . .

Stock keeping records:

- They must contain the quantity of stock on hand; the quantity of losses; and the quantity of adjustments, by individual product.
- The pharmacist and other pharmacy staff are responsible for completing.
- They are recorded on the stock keeping record whenever products are received or issued.
- Entries are also recorded when stock is counted during a physical inventory, or as soon as a loss is noticed.
- When the stock keeping record is full, a new record is started, using the ending balance from the previous record.
- The most common formats for stock keeping records are individual stock cards and stores ledgers. Types of stock keeping records include **stock record cards and bin cards**.

Cont . . .

Stock Bin card

- It is an individual stock keeping record that holds information about a single product by lot number or batch number.
- Every item in that lot will have the same expiration date.
- For example, one bin card would hold information about a single lot of paracetamol at a storage facility.
- The card should note the stock on hand of paracetamol for that lot only, as well as any losses and adjustments for that lot.
- Bin cards are usually displayed at the bins (or shelf or pallet position) where the lot is found.

Cont . . .

Stock record card

- It is an individual stock keeping record that holds information about **all the lots** of a single product.
- one stock record cards is kept for each product.
- The stock record card may be a summary of many bin cards for a particular product.
- For example, one stock record card could hold information about all the paracetamol in a storage facility.

Cont . . .

Transaction records

- Transaction records are used to record information about the movement of stock from one storage facility to another and from storage facility to health facility pharmacies and service delivery points.
- In addition, transaction records are proof of requisition, issue, and/or delivery.
- Warehouse personnel at both issuing and receiving facilities complete transaction records.
- In pharmacies or SDPs, storekeepers, pharmacy personnel, or nurses may complete the transaction records.
- The most common formats are **issue vouchers (model 22); receipt vouchers (model 19) and report and requisition forms (RRF, HPMRR), and internal facility report and requisition forms(IFRR).**

Cont . . .

Determination of stock levels

- The main types of stock levels are:
 - Minimum/safety/reserve stock level
 - Reorder stock level
 - Maximum stock level

Cont . . .

Minimum stock level

- Is the minimum quantity that should always be available
- Is used to protect shortages which could occur when
 - Deliveries are delayed
 - Working stock is consumed at an unexpectedly high rate/increasing demand
- It is determined based on the lead time

Cont . . .

Reorder stock level

- Is the level of stock at which new orders should be placed to replenish the supply/stock.
- $\text{Reorder stock level} = \text{Safety Stock} + (\text{Lead time} \times \text{average monthly consumption})$

Cont . . .

Maximum stock level

- Is the maximum amount of drug items that are allowed to keep in the store at any time.
- $\text{Maximum stock level} = \text{Reorder stock level} + \text{amount to be ordered}$

Cont . . .

- **Types of Max-Min Inventory Control Systems**

Three Types of Max-Min Inventory Control Systems

The same formula is used to determine how much to order or issue for each of the systems

The basic difference b/n the systems is the **trigger** for ordering or issuing

In a **forced-ordering system**

- ☞ end of the review period

In a **continuous review system**

- ☞ when the facility reaches the minimum level.

In a **standard system**

- ☞ end of the review period for the commodities that are at the minimum level.

Cont . . .

Determining How Much to Order or Issue

Stock Status : Purpose of assessing stock status is to determine how long supplies will last.

$$\text{Months of stock on hand} = \frac{\text{stock on hand}}{\text{AMC}}$$

Cont . . .

Calculating Order or Issue Quantities

Order/issue quantity = Max stock quantity – stock on hand

If a facility, for some reason, has not received the previous order or issue, but is positive that stock will arrive (pipeline stock), they should subtract the quantities expected from the next resupply quantity.

Order/issue quantity = Max stock quantity – stock on hand – quantity on order

Cont . . .

1. Forced-ordering max-min system

The inventory control system of **Ethiopia**.

all facilities are required to report on a fixed schedule for all products

Health Posts – **monthly**

Health Centers and Hospitals – **every other month**

Storekeeper decision rule to resupply

At the end of each review period, review all stock levels and order or issue enough stock to bring the levels up to the **max**.

Place an emergency order if the stock level for any item falls below the emergency order point before the end of the review period.

all products are re-supplied each time a report is completed

Level	Review Period	Maximum Months of Stock	Minimum Months of Stock	Emergency Order Point
Health Centres and Hospitals	Every other month	4 months	2 months	0.5 months (= 2 weeks)
Health Posts	Monthly	2 months	1 months	0.25 months (= 1 week)

Cont. . .

Advantages of forced-ordering max-min system

- ✓ rule for resupply is **simple**
- ✓ order/issue every item at the **end of the period.**
- ✓ easier to ensure the availability of transport resources b/c orders are placed at regular intervals
- ✓ Every facility orders or is **resupplied at the end of every review period**
- ✓ **No need of storekeepers constantly assess stock status, unless they think a potential stock out is possible**

Disadvantage of a forced-ordering system

- ✓ orders for some items may be for small quantities
- ✓ B/c all items are ordered regardless of the stock on hand.

Cont. . .

- **Forced –ordering variation: delivery truck system**

also called a **topping up** or **bread truck system** or **Vendor Managed Inventory (VMI) system**.

–difference b/n **regular vs delivery truck forced-ordering system** the way **the deliveries are made** a truck is loaded with supplies at the end of the review period.

The truck and a delivery team travel to each facility assess the stock & leave (top up) an amount of each product

Advantages of the forced-ordering delivery truck system

- ✓ The order is filled on the spot
- ✓ the lead time is zero → lead time stock to zero
- ✓ lowers the min and the max stock levels.
- ✓ Damaged or expired products can be put back on the truck for disposal

Cont. . .

- ✓ The truck can be sent out with a full load of supplies, eliminating multiple small orders.
- ✓ The LMIS report can be completed and collected at the time of delivery.
- ✓ Avoid delayed reporting
- ✓ The training requirements are significantly less for the facility staff.
- ✓ If a supervisor goes on the truck for deliveries, on-the-job training and supervision at the various stops.

Disadvantages of the forced-ordering delivery truck system

- ✓ All types of max-min systems rely on their delivery trucks.
- ✓ system is vulnerable to breakdowns

Cont. . .

Disadvantages of the forced-ordering delivery truck system

- ✓ All types of max-min systems **rely on their delivery trucks.**
- ✓ system is vulnerable to breakdowns
- ✓ A sufficient number of staff must be available in the office
- ✓ team leaders are away making deliveries.
- ✓ Require larger trucks
- ✓ always carry more stock than they will actually deliver.

Cont. . .

Continuous review max-min system

If the stock level is at the min, or has fallen below the min, order enough stock to bring the

In a continuous review system—

- ✓ The review period is **not fixed**
- ✓ The storekeeper **must know both the max and min stock levels.**
- ✓ The storekeeper does **not need an emergency order point**
- ✓ an order can be placed any time stock is needed.
- ✓ The **storekeeper must assess stock status each time an issue is made.**
- ✓ storekeeper's **workload increases if there is many item**
- ✓ The storekeeper must be able to order – pull system
- ✓ b/c only the storekeeper determine stock level (min or not)

A continuous review system must be a **pull system.**

Cont. . .

Advantages of continuous review max-min system

- ✓ rule for resupply is simple: reaching minimum
- ✓ The system is more responsive and flexible
- ✓ orders can be placed at any time.
- ✓ Small orders are eliminated
- ✓ stock levels are at the min when an order is placed.

Disadvantages of a continuous review system

- ✓ Transportation resources are harder to schedule.
- ✓ Overload storekeepers in large facilities with a large number of products

Cont. . .

- **Continuous review system variation: two bin**

Resupply rule is the same

The difference the way of determining when the min has been reached.

two equal-sized bins of each **individual product**

When the **first bin is empty**, the min has been reached.

Cont. . .

Advantages continuous review system variation: two bin.

- ✓ Require less training than a normal pull systems
- ✓ No calculations are required and paperwork is minimal.

Disadvantages:

- ✓ bin size must be continually reviewed if consumption for products is not stable

Cont. . .

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- ✓ bin size must be continually reviewed if consumption for products is not stable

Cont. . .

Standard max-min system

Resupply (when to and how much to order) is based on

- Reviewing all stock levels at the end of each review period.
- Falling below the min
- In a standard system—
 - Ordering/issuing new stock is based on the min stock level and the review period.
 - the storekeeper must know the min, max, and review period.
 - an emergency order point is needed to ensure that a stockout does not occur between review periods.
 - the stock status must be assessed at the end of each review period

Cont. . .

Advantages of standard max-min system

- ✓ Eliminate small orders
- ✓ Avoid the need to assess stock status continually
- ✓ Reduce the number of calculations
- ✓ Transportation can be scheduled for specific times

Disadvantages:

- ✓ Higher min stock level
- ✓ Increase the likelihood of expiry
- ✓ Require more storage capacity,
- ✓ Increased costs.
- ✓ More training for the storekeepers may be required b/c of their decision rules are more complex.

Characteristics of a well managed stores operation

- Store divided into zones
- Stocks arranged within each zone in some order system
- Stocks stored on pallets on the floor or on shelves
- Store products based on **manufacturer' specification**
- **Liquid and heavy products** kept at the lower shelves
- Products arranged to facilitate **first expiry first out**
- Damaged/expired products stored separately
- Report products with near expiry
- Good housekeeping- cleaning and inspection
- Clearly defined management structure
- Staffs appropriately qualified, trained, disciplined and rewarded

THANK YOU!!